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**Saveetha Institute of Medical and Technical Sciences ,
Chennai-602105**

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Secure File Encryption and Decryption System with Password Authentication, Advanced Data Protection, and Unauthorized Access Prevention

**COURSE NAME:C Programming
COURSE CODE:CSA0206**

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Introduction:

- Data security is a major concern in the digital world.
- Unauthorized access to files can lead to data theft and privacy issues.
- Encryption converts readable data into unreadable format.
- This project provides secure file protection using encryption and password authentication.
- The system is developed using C programming.



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Problem Statement:

- Sensitive files are vulnerable to unauthorized access.
- Traditional file storage like Folder-Based Storage System methods lack proper security.
- Data breaches can cause financial and personal loss.
- There is a need for a secure and lightweight file protection system.



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Objectives:

- To develop a secure file encryption system.
- To provide password-based authentication.
- To protect confidential files from unauthorized users.
- To implement encryption and decryption operations.
- To ensure data confidentiality and integrity.



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Proposed System:

- The system encrypts files using encryption algorithms.
- Only authenticated users can decrypt files.
- Password verification ensures secure access.
- The application supports secure storage of sensitive data.
- Menu-driven interface improves usability.

Modules of the System:

Modules:

1. User Authentication Module
2. Encryption Module
3. Decryption Module

Technologies Used:

- C Programming
- File Handling

Module 1 :User Authentication Module

Key Features

- Secure user registration
- Username and password-based login
- Password encryption and secure storage
- User identity verification
- Unauthorized access prevention
- Session management and secure logout

Working Process

- User enters username and password.
- System validates the credentials.
- If authentication is successful, access is granted.

If authentication fails, an error message is displayed and access is denied.

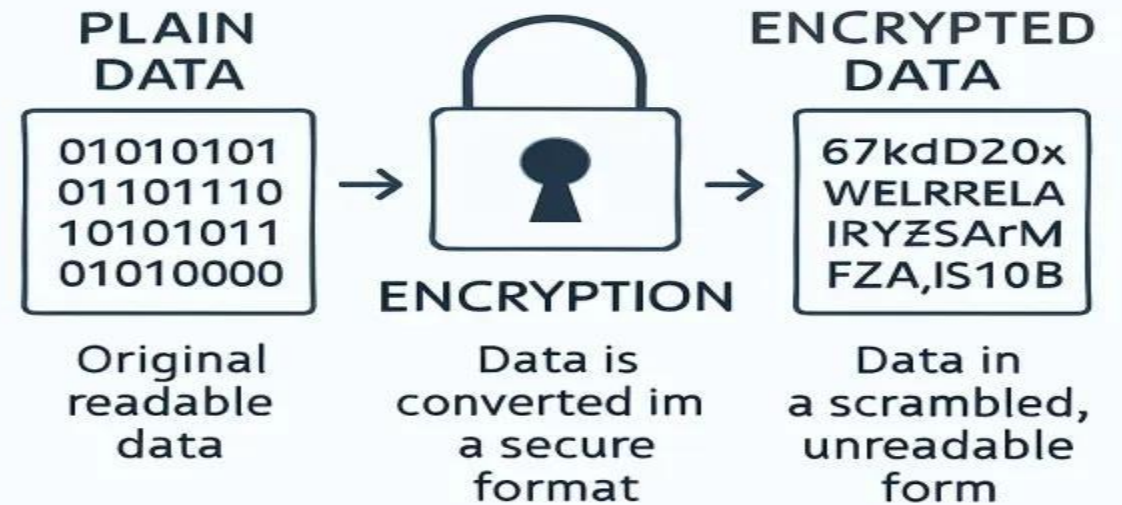
Module 2 Encryption

Process:

Steps:

1. User selects file
2. Password is entered
3. Encryption algorithm processes data
4. File content becomes encrypted
5. Encrypted file is stored securely

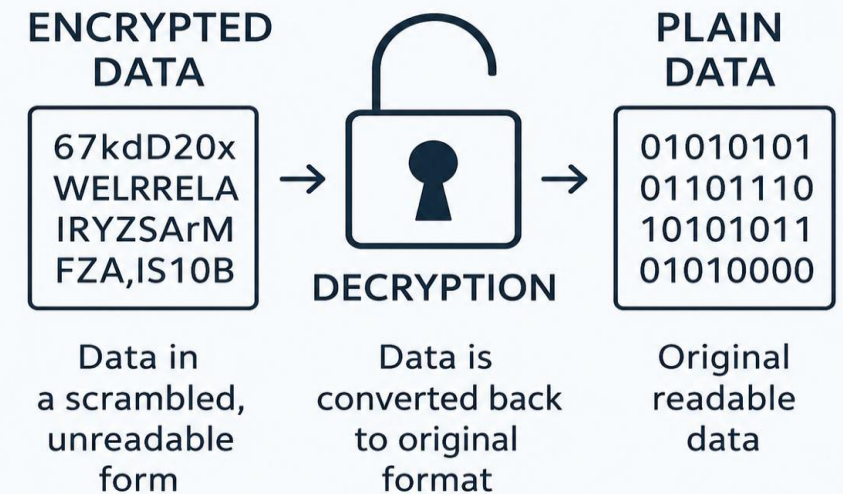
HOW DOES ENCRYPTION WORK?



Module 3 Decryption Process: Steps

1. User enters correct password
2. System verifies authentication
3. Encrypted file is accessed
4. Decryption algorithm restores original data
5. Original file is recovered

HOW DOES DECRYPTION WORK?



Title	Author	Year Published	Improvement in Our Project
Secure Data Encryption Through a Combination of AES, RSA and HMAC	E. S. I. Harba	2017	Our system provides a simpler password-based file encryption system with an easy-to-use interface while maintaining strong security.
A Secure File Encryption and Decryption System Using AES for Text and Images	Pranav Banga, Nirottam, Shubham Kumar Singh,	2023	Our project supports secure encryption and decryption of different file types with password authentication and unauthorized access prevention.
AES-RSA: An Innovative Hybrid Security Framework for File Authentication, Integrity, and Data Secrecy Model	Smita Rath et al.	2024	Our system is lightweight, faster, and easier to deploy while providing strong file protection through password authentication.
Hybrid Cryptographic System for Secure, Scalable File Encryption and Decryption Using RSA and AES	Mohit Kumar Malviya, Manish Kumar Singhal	2024	Our project focuses on secure local file encryption with a simple password-based approach suitable for personal and educational use.



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Conclusion:

- The project provides secure protection for digital files.
- Encryption ensures confidentiality of data.
- Password authentication prevents unauthorized access.
- The system demonstrates practical cybersecurity concepts using C programming.
- It offers a reliable and efficient data security solution.



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- Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Pearson Education.
- William Stallings, Cryptography and Network Security, Pearson Education.
- Behrouz A. Forouzan, Cryptography and Network Security, McGraw Hill.
- Yashavant Kanetkar, Let Us C, BPB Publications.



Thank You